

Separating Neighboring River Plumes Using End-Member Mixing Analysis

Winyah Bay and Santee River plumes | March 2025 observations
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Main Point: Neighboring river plumes can overlap, making their freshwater sources difficult to distinguish from standard hydrographic observations. Using plume-layer salinity and CDOM in a three-endmember mixing analysis, we estimated Winyah Bay, Santee River, and ocean contributions at each sampled station. The resulting patterns were spatially consistent and mostly changed with expected wind-driven plume movement.

1. Problem & Field Data

Can commonly measured water properties distinguish contributions from two adjacent, interacting river plumes?

Background and Motivation
River plumes transport freshwater, sediments, nutrients, and pollutants onto the coastal shelf while modifying stratification and circulation (Horner-Devine et al. 2009, 2015). When neighboring plumes overlap, salinity alone may no longer identify the source of the water. Warrick and Farnsworth (2017) showed that plume collisions and coalescence can be common, highlighting the need for observational methods that distinguish contributions from nearby rivers.

The Winyah Bay and Santee River Study Area
Winyah Bay and the Santee River discharge into the South Atlantic Bight through outlets less than 15 km apart. Winyah Bay has an estimated average freshwater discharge of approximately 557 m³ s⁻¹ (Kim and Voulgaris 2008; Patchineelam 1999), while the Santee River averages approximately 311 m³ s⁻¹ (Hockensmith 2004). Their proximity allows the two plumes to overlap and mix across the shallow continental shelf.

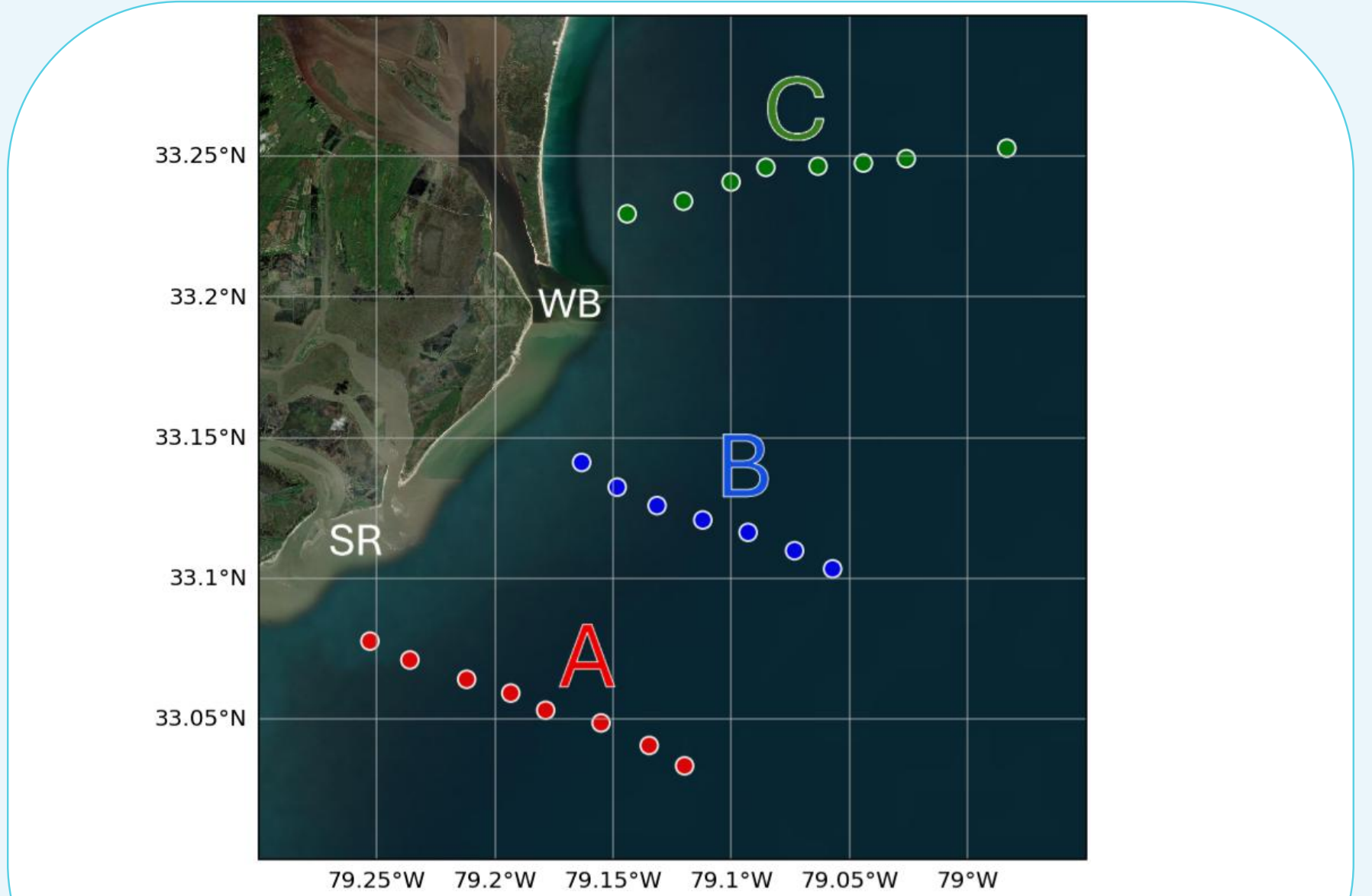


Figure 1. March 2025 study area and CTD transects. Transects A, B, and C span the region between the Santee River (SR) and Winyah Bay (WB), with transect B located where plume interaction was expected.

Field Observations and Sampling Design

- Data were collected aboard the R/V Savannah from March 9-13, 2025.
- 63 CTD stations were sampled along three cross-shelf transects between the two river outlets.
- Transects were sampled repeatedly to capture changes in plume structure through time.
- CTD observations provided vertical profiles of salinity, density, CDOM, and other supporting water properties.
- Shipboard wind measurements were used to interpret changes in plume position and source influence.

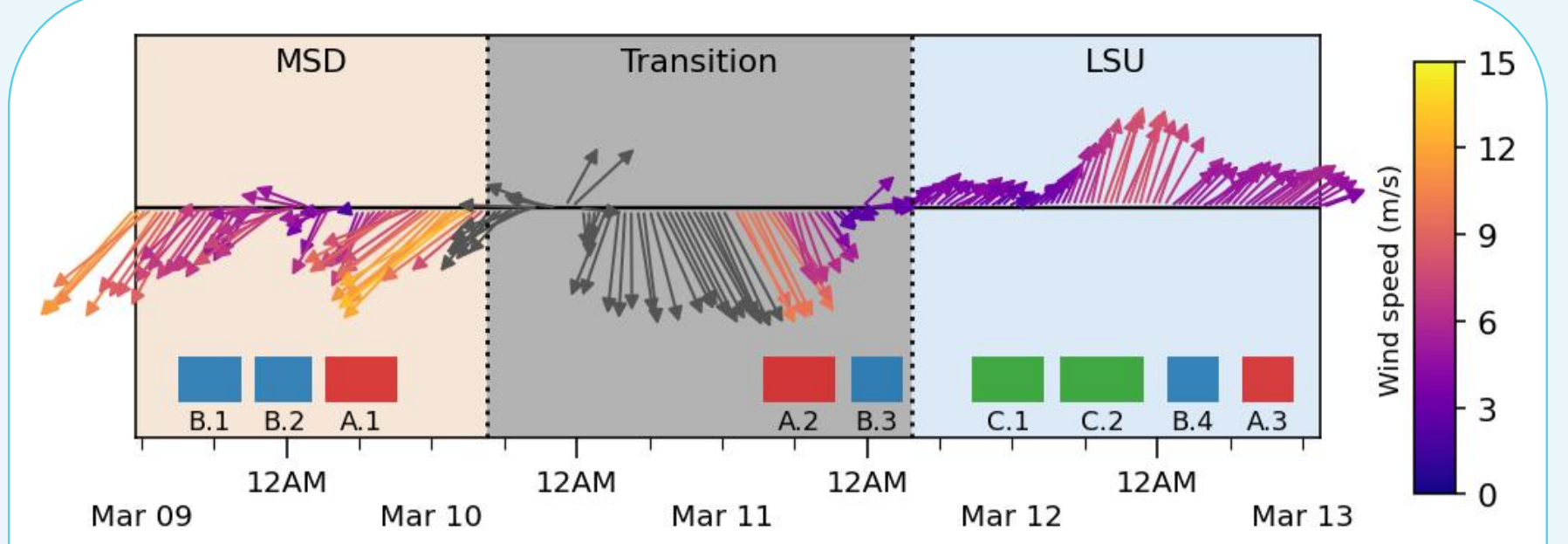
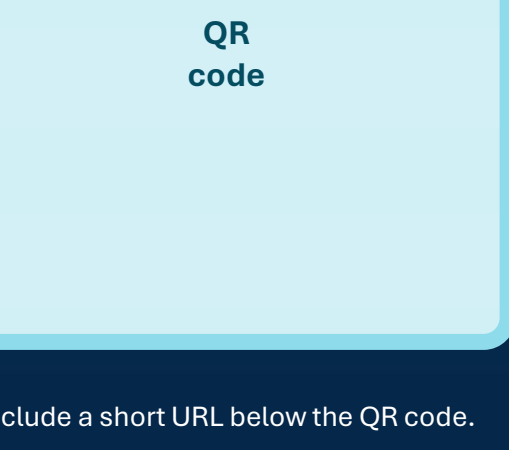


Figure 2. Shipboard wind vectors during the cruise. The sampling periods occurred across moderately strong downwelling-favorable winds (MSD), a transition period, and low-strength upwelling-favorable winds (LSU).

Full Paper, References, Supplementary Figures, and Satellite Animations



Include a short URL below the QR code.

2. EMMA Method

Plume-Layer Averaging
A density-based plume depth was calculated for each CTD profile to identify the portion of the water column influenced by buoyant freshwater (Arneborg et al. 2007; Papageorgiou 2023). Salinity, CDOM, and the other measured properties were then averaged between the shallowest observation and the estimated plume depth. Using plume-layer averages allowed stations with different water depths and vertical structures to be compared consistently.

Tracer Selection
Candidate tracers were evaluated using the criteria described by Cuoco et al. (2021): Effective tracers should show approximately linear mixing behavior, remain reasonably conservative across the study region, differ among the source waters, and have stable endmember values during the sampling period.

Among the available CTD variables, salinity and CDOM showed the clearest near-linear relationship and the strongest potential to distinguish the freshwater sources. Variables affected more strongly by biological activity or air-sea exchange were considered less reliable for the mixing analysis.

Endmember Selection
The model included three source waters: Winyah Bay (WB), Santee River (SR), and Ocean (OC). Freshwater endmembers were selected from low-salinity stations near each plume source. The ocean endmember was selected from a high-salinity, weakly stratified offshore station with minimal CDOM.

Three-Endmember Model
For each station, the observed plume-layer salinity and CDOM were represented as a mixture of three endmembers (Christophersen and Hooper 1992; Han et al. 2012; Cuoco et al. 2021):

$$\begin{aligned} f_{WB} S_{WB} + f_{SR} S_{SR} + f_{OC} S_{OC} &= S_{obs} \\ f_{WB} C_{WB} + f_{SR} C_{SR} + f_{OC} C_{OC} &= C_{obs} \\ f_{WB} + f_{SR} + f_{OC} &= 1 \end{aligned}$$

Here, S and C represent plume-layer salinity and CDOM, and each f represents the estimated fraction contributed by Winyah Bay, the Santee River, or the ocean. Solving the system produced three source-water fractions for every station.

Endmember Source-Fractions
The calculated fractions were plotted in ternary space. Each vertex represents one endmember, points along an edge represent mixtures of two sources, and points within the triangle contain contributions from all three. Stations were classified by the endmember with the largest calculated fraction, allowing the source estimates to be compared geographically and across changing wind conditions.

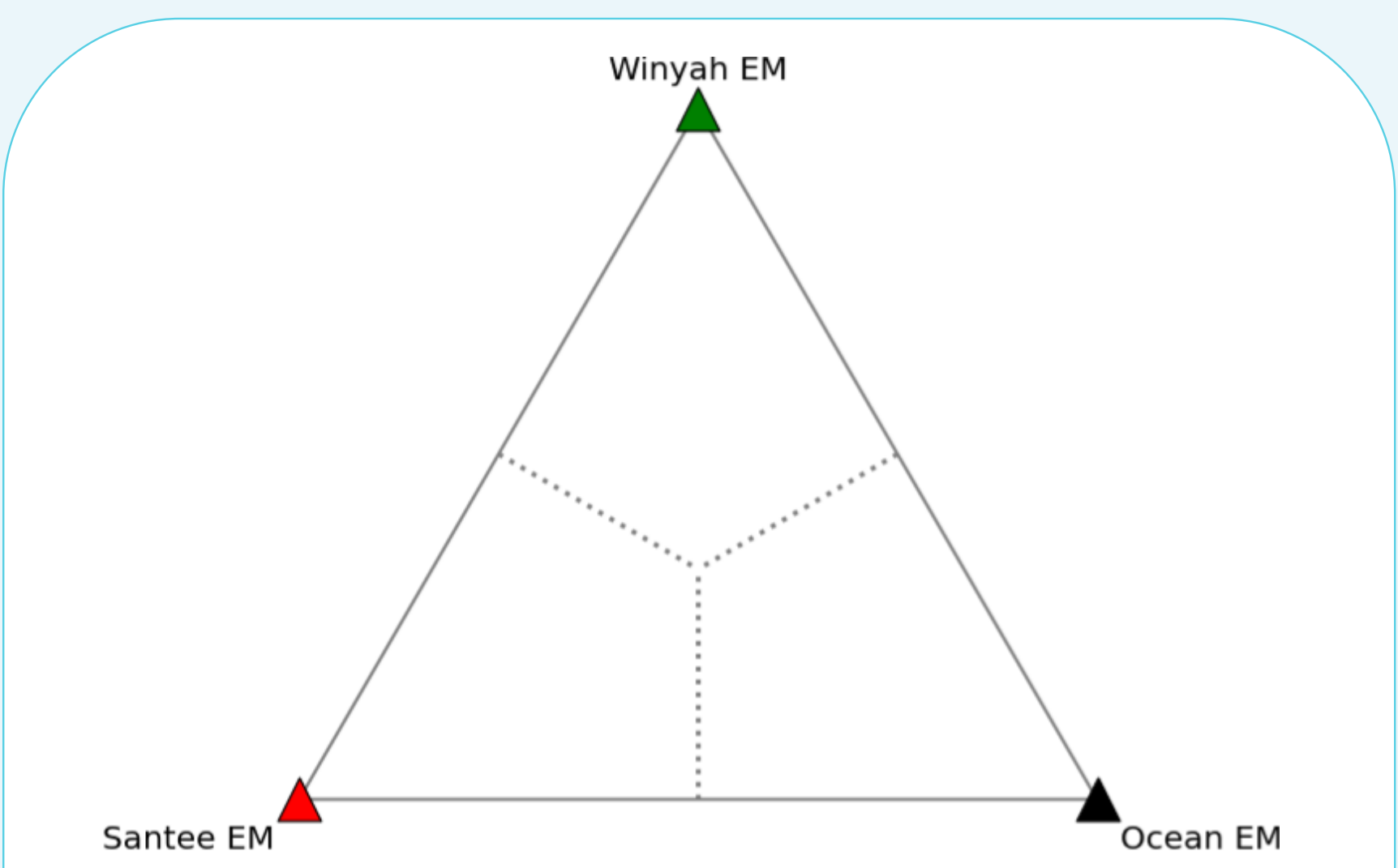


Figure 3. Ternary space for the three-endmember model. The vertices represent pure Winyah Bay, Santee River, and ocean endmembers. A station's position within the triangle is determined by its calculated source fractions; movement toward a vertex indicates increasing influence from that endmember.

3. Main Results

Main finding: EMMA revealed geographically consistent Winyah Bay, Santee River, and ocean contributions where standard property visualizations showed substantial overlap. The resulting source patterns changed consistently with the observed wind forcings.

Standard Plots: Sources Unresolved
Depth profiles, contour plots, maps, and water property comparisons could clearly distinguish plume-influenced water from more oceanic water. However, measurements from the three transects overlapped strongly, so no single property or simple threshold could reliably determine whether a station was influenced more by Winyah Bay or the Santee River.

EMMA: Plume Sources Resolved
The EMMA converted the overlapping salinity and CDOM observations into fractional contributions from the three source waters. In ternary space, stations generally grouped according to their geographic setting: Transect C stations clustered toward the Winyah Bay-ocean side, while Transect A stations clustered toward the Santee River-ocean side. Transect B showed influence from both plume sources.

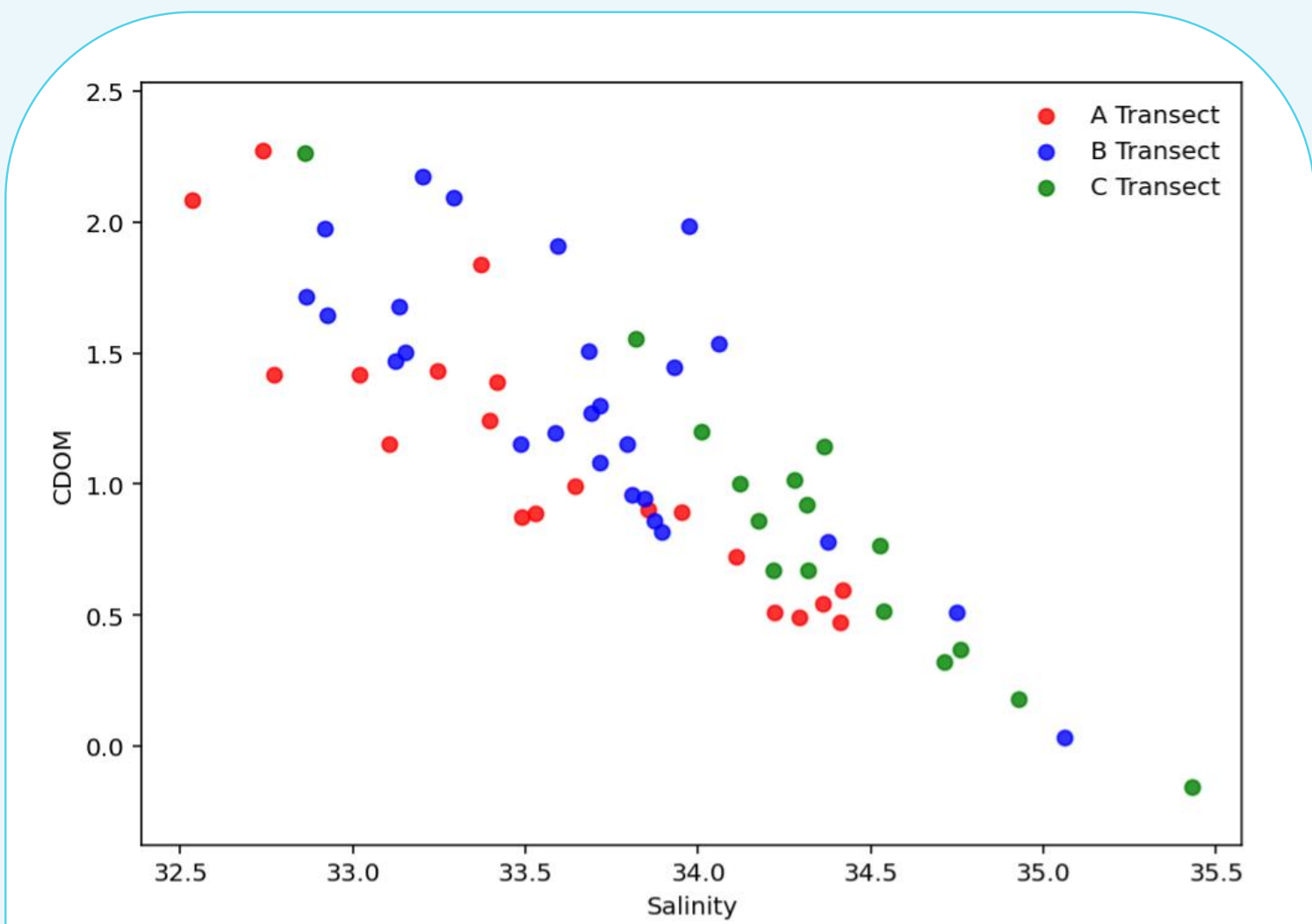


Figure 4. Plume-layer averaged salinity versus CDOM. This tracer pair shows the clearest structure and grouping, but transects still overlap, motivating a mixing analysis rather than a simple visualization.

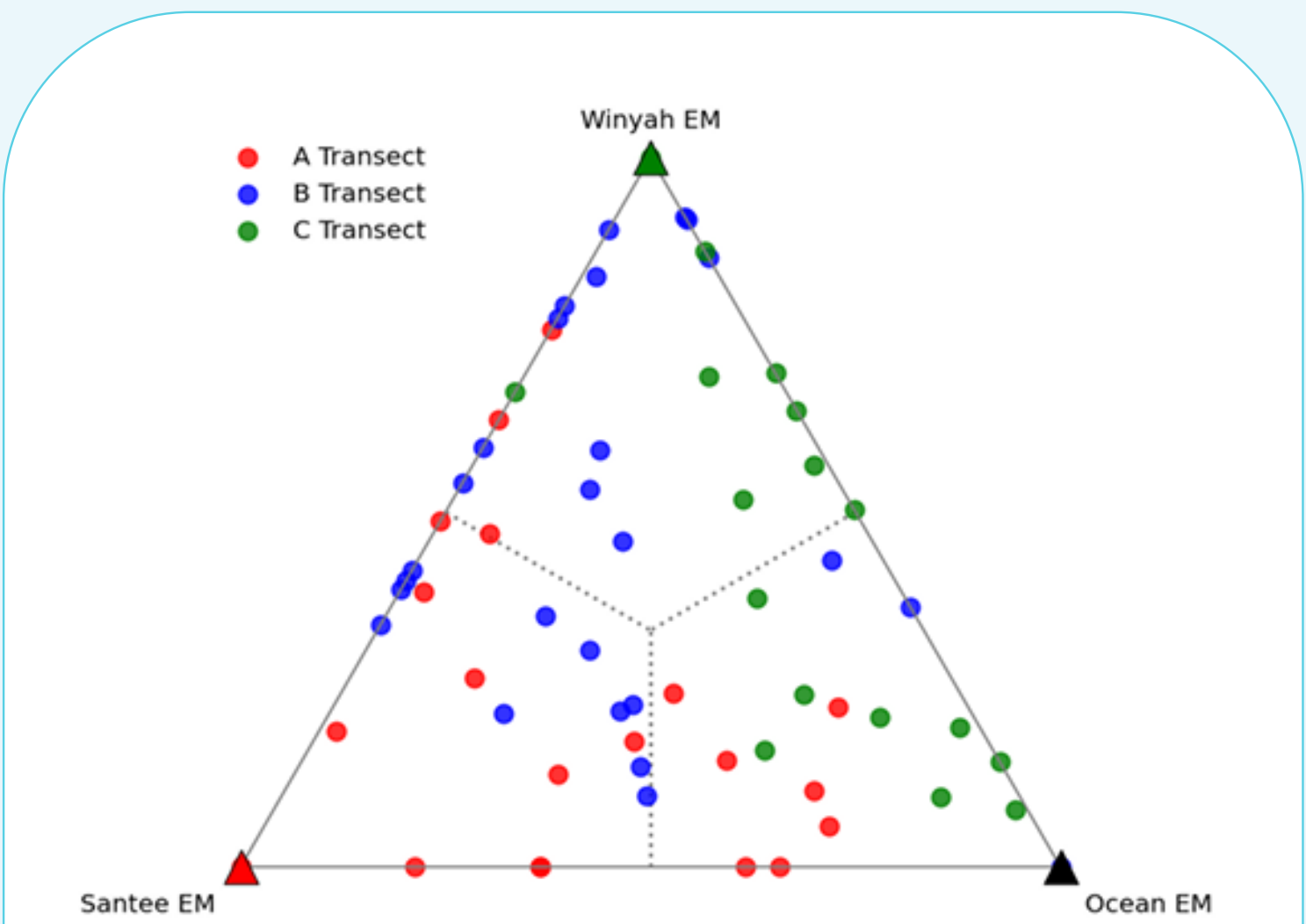


Figure 5. EMMA-derived source fractions. Transect C clusters toward the Winyah-ocean side of ternary space, while transect A clusters toward the Santee-ocean side.

Source Dominance Shifted with Wind Forcing
The EMMA fractions were used to classify each station according to its largest endmember contribution: Winyah Bay, Santee River, or ocean. During the moderately strong downwelling-favorable regime, landward stations were generally Winyah-dominant, while more offshore stations tended to be Santee- or ocean-dominant. During the transition period, Transect B remained predominantly Winyah-dominant, while Transect A shifted from Santee dominance nearshore to ocean dominance offshore.

These patterns are consistent with downwelling-favorable winds compressing buoyant plumes toward the coast and promoting down-coast, along-shelf transport (Dykstra et al. 2024; Horner-Devine et al. 2009). Under the weaker upwelling-favorable regime, Transects A and B became nearly entirely Santee-dominant, while Transect C transitioned from Winyah dominance nearshore to ocean dominance offshore. This increased Santee influence across the northern and central study region is consistent with upwelling-favorable winds driving plume water offshore and toward the northeast (Yankovsky et al. 2022). Overall, the changing dominance patterns broadly matched the expected wind-driven movement of the two plumes.

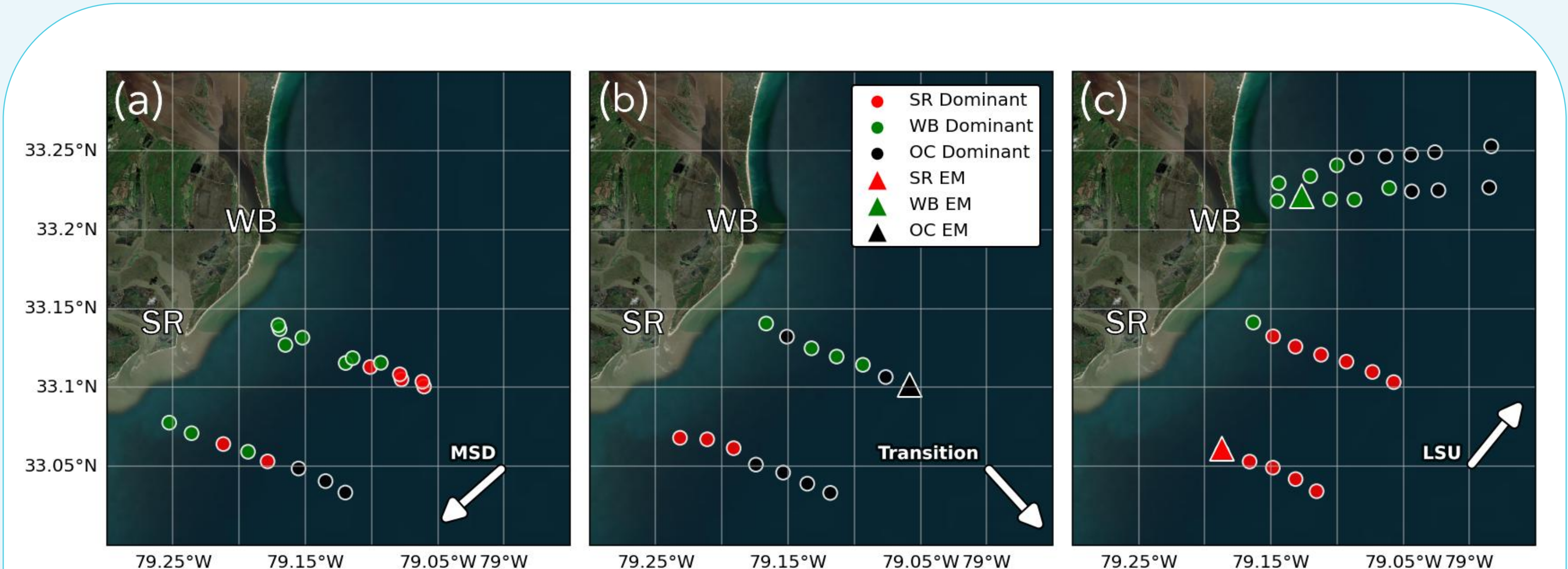


Figure 6. Geographic classification maps grouped by wind regime: (a) Medium Strength Downwelling (MSD), (b) Transition, and (c) Low Strength Upwelling (LSU). An arrow on each map shows the mean wind direction during that regime. Each station was classified by the endmember contributing its largest EMMA-derived fraction. The maps show how dominant Winyah Bay, Santee River, and ocean influence changed across the three wind regimes.

4. Conclusions & Details

Conclusions
Standard plume visualizations identified plume-influenced water but could not reliably distinguish Winyah Bay from Santee River influence. By combining plume-layer salinity and CDOM, EMMA converted overlapping observations into quantitative Winyah Bay, Santee River, and ocean fractions. These fractions formed geographically consistent patterns that varied consistently with the observed wind regimes.

Limitations
The EMMA method relies on several assumptions and limitations. A primary limitation is the assumption that tracer values for each freshwater endmember remain effectively constant during the study period. If discharge and tracer concentrations changed between fixed endmember characterization and plume sampling, the calculated fractions may be biased toward the endmember whose tracer values most closely match the shifted plume properties.

In this dataset, there was a tradeoff between selecting stations representative of plume conditions during the cruise and stations closest to unmixed source waters. Because the endmembers were inferred near the plume origins rather than sampled directly within the estuaries, they likely already incorporated early mixing and transformation with ambient shelf waters. This uncertainty may have contributed to several offshore stations being classified as Santee-dominant during downwelling-favorable conditions. The results are therefore best interpreted as regional patterns rather than exact source assignments for individual stations.

Future Work
Future applications should directly sample source waters within each estuary and pair these observations with near-field plume sampling. Endmembers should also be sampled repeatedly through time, particularly during changing discharge conditions, to determine how variability in source-water properties affects the calculated fractions.

Applying EMMA to other coastal systems would require representative endmembers, conservative or near-conservative tracers that differ among the sources, and sampling that covers the full region of plume overlap. With more tracers, the framework could include additional rivers or other water masses, such as upwelled water.

Broader Significance
This observational EMMA provides quantitative source-water estimates that could support the interpretation and evaluation of interacting-plume models. When combined with physical and biogeochemical observations, EMMA could provide a bridge between satellite imagery, which maps surface plume extent, and numerical models, which simulate plume dynamics and interactions, and together offer a practical method for future observational studies in other multi-plume coastal environments.

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